

Draw Back in Static Routing

- \* Time Delay
- \* Human error

Dynamic Routing Protocol:-

It is a protocol  
in use to Exchange the Route Automatically  
To your Next Router

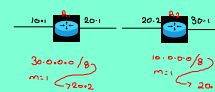
RIP }  
EIGRP } Internal Gateway Protocol  
OSPF }  
BGP } External Gateway Protocol

### RIP

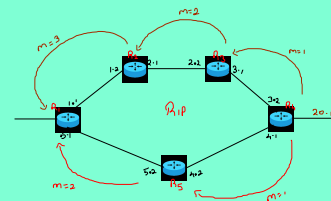
Routing Information Protocol  
It is one of dynamic routing protocol  
It will send update packet as broadcast 255.255.255.255  
It is work with help of UDP protocol  
Metric = hopcount  
Maximum metric = 15  
If M=16 is announce as invalid or poison route



```
R1(config)#router rip
R1(config-router)#network 10.0.0.0
R1(config-router)#network 20.0.0.0
```



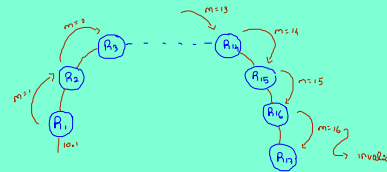
update  
30.0.0.0/8  
m=1  
SIP: 20.0.0.2  
DIP: 255.255.255.255



20.0.0.0/8  
m=3  
10.0.0.2

20.0.0.0/8  
m=2  
50.0.0.2

Lowest metric will  
Select as Best Route



## EIGRP

Enhanced Interior Gateway Routing Protocol

Advanced Distance Vector Routing Protocol

It contains 3 tables

- \* Neighbor table      Next Router info
- \* Topology table      Best Route / Backup
- \* Routing              Best Route

metric: composite parameters / K-values

|                |             |
|----------------|-------------|
| K <sub>1</sub> | Bandwidth   |
| K <sub>2</sub> | Load        |
| K <sub>3</sub> | Delay       |
| K <sub>4</sub> | Reliability |
| K <sub>5</sub> | MTU         |

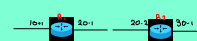
| K1         | K2         | K3           | K4 K5       |
|------------|------------|--------------|-------------|
| Bandwidth  | Load       | Delay        | Reliability |
| $K1 = BW$  | $K2 = BW$  | $K3 = DLY$   | $K4 = REL$  |
| $256 / BW$ | $256 / BW$ | $10^7 / DLY$ | $256 / REL$ |
| 0          | 0          | DLY          | 1           |

Default K-values :-

- K<sub>1</sub> = 1
- K<sub>2</sub> = 0
- K<sub>3</sub> = 1
- K<sub>4</sub> = 0
- K<sub>5</sub> = 0

$$256 \left\{ \frac{10^7}{\text{Least Bandwidth in Route}} + \text{total Delay} \right\}$$

```
R1(config)#router eigrp 10
R1(config-router)#network 10.0.0.0
R1(config-router)#network 20.0.0.0
```



Neighbor Ship:

By Sending Hello Packet Automatically

Hello
Success
 Src: 10.0.0.10 → Dest: 224.0.0.10 → Multicast IP

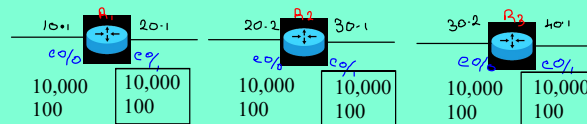
## Neighborship condition

- \* Same Network
- \* Same ASN
- \* Same K-value

## EIGRP timer

Hello time = 5 sec

Hold on time = 15 sec



R1#show interface eth0/0

$$256 \left\{ \frac{10^7}{\text{least Bw in Route}} + \text{total Delay} \right\}$$

$$256 \left\{ \frac{10^7}{10,000} + 300 \right\}$$

$$256 \{ 1000 + 300 \}$$

$$256 \{ 1300 \}$$

$$= 332800$$

## OSPF

Open shortest path first  
 It is open standard protocol  
 It is link state routing protocol  
 Rip & eigrp is distance vector routing protocol  
 It will send hello packet using 224.0.0.5 multicast ip  
 To avoid broadcast issue in BMA network we have DR BDR DR-O  
 To send update packet to DR BDR It will use 224.0.0.6  
 It contains three table  
 • Neighbor table — It contains neighbor information  
 • Link state table — It contains map info  
 • Routing table — It contains only best route  
 Metric = COST  
 Metric formula  

$$\text{Cost} = \frac{10^8}{\text{bandwidth in bits}}$$

### Router ID :

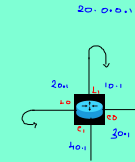
Identification of OSPF Router

IP format x.x.x.x

manually assign by Admin

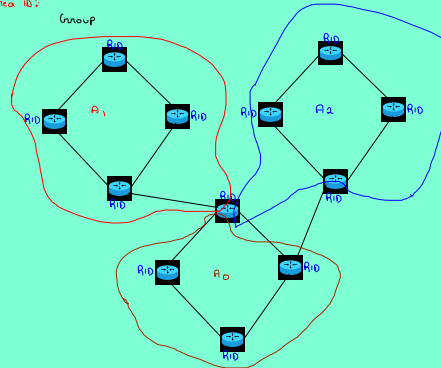
Highest IP in Loop back int

Highest IP in Physical int



### Area ID:

Group



### WCB

used and bits

0 - care

1 - don't

10.0.0.0

0.255.255.255

matching

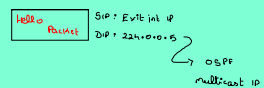
Net mask

1 - care

0 - don't

### Neighbor ship

By Sending Hello Packet



Neighbor Ship condition:

Same Network

Same Area

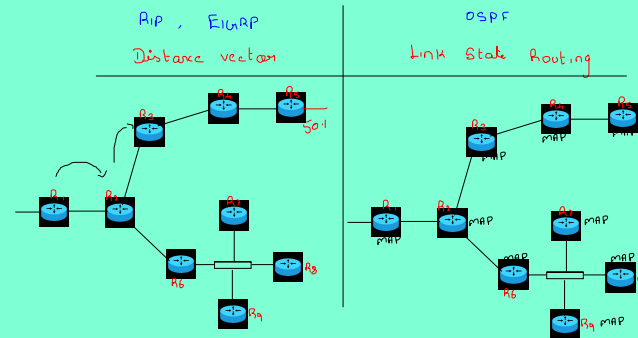
unique Router id

Same time

ospf timer

Hello time : 10 Sec

Dead time :  $4 \times 10 = 40 \text{ Sec}$



Cost  $\rightarrow$  Interface Bandwidth

